HYDROTECH – Technical Progress Sheet

1. Problem

Traditional water meters mainly measure consumption but lack real-time monitoring, automated billing, water-quality monitoring, leak/theft detection, and remote valve control. Water-service providers also face challenges with manual operations, billing errors, and disconnected records.

2. Proposed Solution

Smart H_2O is a self-powered IoT-based smart water monitoring system that provides real-time consumption and water-quality monitoring, automated billing, leak/theft detection, and remote valve control. Users can monitor their usage through a mobile app, while authorities can manage services through a web dashboard.

3. Architecture

Sensors collect water-flow and quality data and send it to the **ESP32**, which processes the data before storing it in the **cloud database**. Users and authorities can access the information through the **mobile app and web dashboard**, while the system enables remote water-valve control.

4. Current Progress

- Smart H_2O prototype has been developed.
- System architecture has been defined.
- Web dashboard has been developed.
- Mobile application interface has been developed.
- Water-valve control is currently in progress.

5. Technology Stack

Hardware: ESP32, flow and quality sensors, water valve, self-power unit
Software: Mobile app and web dashboard
Backend: Cloud database and data processing
Communication: Wi-Fi/IoT
Functions: Monitoring, billing, leak/theft detection, and remote shut-off

6. Testing

Initial laboratory testing is being conducted to validate the prototype. A **10-house pilot test** is planned for real-world evaluation.

7. Limitations

- Further real-world validation is required.
- Self-powered operation needs testing under different flow conditions.
- Remote valve reliability requires further testing.
- Large-scale deployment needs further cost, connectivity, and maintenance evaluation.